Air Quality Multi-Horizon Forecasting

Final Report

Overall Forecasting Framework

The proposed framework uses a **horizon-adaptive, pollutant-specific strategy**:

- **Short-term (1–5 days):** Deep Learning / ML models capture short-term spatiotemporal variations.
- **Long-term (150 days):** Low-parameter statistical, climate-informed, and hazard-classification models are used according to pollutant characteristics.

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1. Short-Term Forecasting — 1–5 Days

Pollutant	Best Model	Approach	R^2	Pearson r	SSIM	CSI / AUC
NO_2	**ConvGRU Decoupled ST-ResUNet**	Deep Learning	**0.651**	**0.810**	**0.896**	CSI = 0.361
CO	**Multi-Scale Regularized Ridge**	ML	**0.693**	**0.835**	**0.719**	CSI = 0.091
SO_2	**Two-Stage Hurdle Model**	ML / Hazard	**N/A**	0.275	0.256	**AUC = 0.616**

Short-term interpretation

- **NO_2:** Strong spatiotemporal performance with **R^2 = 0.651** and **SSIM = 0.896**.
- **CO: Multi-Scale Regularized Ridge achieves the highest short-term R^2 = 0.693**.
- **SO_2:** Continuous regression is less suitable; the Hurdle model predicts **plume occurrence/risk** instead of exact concentration.

2. Long-Term Forecasting — 150 Days

Pollutant	Champion Model	Approach	R^2	Pearson r / AUC	Other Metric
NO_2	**Monthly Spatial Climatology**	Statistical	**0.601**	**r = 0.740**	SSIM = 0.852
CO	**Harmonic + ENSO/IOD**	Statistical + Climate	**0.683**	**r = 0.836**	SSIM = 0.711
SO_2	**HistGradientBoosting**	Hazard Classification	**N/A**	**ROC-AUC = 0.634**	PR-AUC = 0.411, CSI = 0.261

Long-term interpretation

- **NO_2:** Monthly climatology provides strong and stable long-range prediction (**R^2 = 0.601**).
- **CO:** Harmonic regression with ENSO/IOD provides the strongest reported long-range continuous prediction (**R^2 = 0.683**).
- **SO_2:** The problem is reformulated as **exceedance/hazard forecasting** because continuous regression performs poorly for the highly sparse pollutant.

3. Final Multi-Horizon Comparison

Pollutant	1–5 Days — R^2	150 Days — R^2	Short-Term Model	Long-Term Model
NO_2	**0.651**	**0.601**	ConvGRU ST-ResUNet	Monthly Climatology
CO	**0.693**	**0.683**	Multi-Scale Ridge	Harmonic + ENSO/IOD
SO_2	**N/A***	**N/A***	Hurdle Model	GBDT Hazard Classifier

* SO_2 is evaluated as a **classification/hazard forecasting problem**, so R^2 is not applicable.

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4. Key Scientific Finding

> **The results support a horizon-adaptive forecasting framework rather than a single model for all pollutants and forecast horizons. Deep learning/ML models perform well for short-term prediction, while climatological and climate-informed models provide competitive long-term forecasts. For highly sparse SO_2, hazard classification is more appropriate than continuous concentration regression.**

Best reported R^2 values

Horizon	Pollutant	Best R^2
1–5 days	CO	**0.693**
1–5 days	NO_2	**0.651**
150 days	CO	**0.683**
150 days	NO_2	**0.601**
SO_2	Both horizons	**R^2 not applicable to selected classification approach**